# Introduction to Large Language Models
## Assignment- 11

**TYPE OF QUESTION: MCQ**
**Number of questions: 10** **Total mark: 10 X 1 = 10**

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**QUESTION 1: [1 mark]**

Which statements correctly characterise the DistMult model for knowledge graph completion?

a. Each relation is parameterised by a full D x D matrix, which lets it capture asymmetric relations.

b. The relation is represented by a diagonal matrix, so the score is a multiplicative interaction of the entity embeddings.

c. Because the diagonal form makes the score symmetric in the subject and object, it cannot distinguish a relation from its inverse.

d. DistMult can only be evaluated on knowledge graphs whose relations are all symmetric.

**Correct Answers:** b, c

**Explanation:**

(b) True: DistMult restricts the bilinear form to a diagonal relation matrix, so the score reduces to the element-wise product of the subject, relation and object vectors summed over dimensions.
(c) True: that product is unchanged when the subject and object are swapped, so DistMult assigns the same plausibility to (s, r, o) and (o, r, s). It therefore cannot model relations such as parent_of.
(a) is false and in fact describes the general bilinear model that DistMult simplifies. (d) is false: DistMult is trained and evaluated on ordinary benchmarks such as FB15k-237, it simply performs poorly on their asymmetric relations.

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**QUESTION 2: [1 mark]**

In DistMult the score of a triple is the sum over dimensions of the element-wise product of the subject, relation and object embeddings. Given

s = [1, 2, 0], r = [2, -1, 3], o = [3, 1, 4]

what are the scores of (s, r, o) and of the reversed triple (o, r, s)?

a. 4 and 4

b. 4 and -4

c. 4 and 12

d. 6 and 4

**Correct Answer:** a

**Explanation:**

score(s, r, o) = (1)(2)(3) + (2)(-1)(1) + (0)(3)(4) = 6 - 2 + 0 = 4.
Reversing the triple reorders the three factors in each term but multiplication is commutative, so every term is unchanged and the score is again 4.
This is the symmetry problem in concrete form: no choice of embeddings can make DistMult score a relation higher in one direction than the other.

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**QUESTION 3: [1 mark]**

Which statements about the ComplEx extension of DistMult are true?

a. It moves the embeddings into the complex domain and takes the complex conjugate of the object embedding in the score.

b. It replaces multiplicative interaction with element-wise addition of real-valued vectors.

c. Setting the imaginary part of a relation embedding to zero recovers a symmetric, DistMult-like score.

d. Every entity embedding must have unit norm for the model to be valid.

**Correct Answers:** a, c

**Explanation:**

(a) True: the conjugation is what breaks the symmetry. Swapping the subject and object now changes the sign of the imaginary contribution, so the two directions of a relation can be scored differently.

(c) True: with a purely real relation embedding the imaginary terms cancel and the score becomes symmetric again, which means DistMult is a special case of ComplEx.

(b) is false - the interaction stays multiplicative. (d) is false: normalisation is a common training choice, not a requirement of the model.

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**QUESTION 4: [1 mark]**

What is the main modification SimplE makes to DistMult-style models in order to handle asymmetric relations?

a. It replaces the entity embeddings with fixed random vectors.

b. It gives every entity two embeddings, one for its subject role and one for its object role, and learns an embedding for each inverse relation.

c. It constrains the relation tensor to rank one.

d. It samples negatives for only half of the training triples.

**Correct Answer:** b

**Explanation:**

With separate head and tail embeddings, the vectors used when an entity appears as a subject differ from those used when it appears as an object, so the score is no longer forced to be symmetric. SimplE then ties the two directions together by averaging the score of (s, r, o) with the score of (o, r_inverse, s), which is what keeps the two role embeddings from being learned independently and losing the connection between them.

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**QUESTION 5: [1 mark]**

What is the main advantage of a factorised representation such as DistMult or ComplEx for a large knowledge graph?

a. It forces every relation in the graph to be symmetric.

b. It stores each entity as a one-hot vector, which simplifies nearest-neighbour search.

c. It collapses the graph into a single scalar summary.

d. It replaces the full adjacency tensor with low-rank factors, which cuts the parameter count and lets the model generalise to unseen triples.

**Correct Answer:** d

**Explanation:**

The full tensor over subjects, relations and objects has $|E| \times |R| \times |E|$ entries, which is impossible to store for a graph with millions of entities. Representing entities and relations by low-dimensional vectors reduces this to roughly $(|E| + |R|) \times d$ parameters.

The low-rank constraint is not just a compression trick: it forces the model to share structure across triples, and that sharing is what allows it to score a triple it never saw during training.

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**QUESTION 6: [1 mark]**

How is a 3D knowledge graph tensor of shape |E| x |R| x |E| commonly reshaped into a matrix factorisation problem?

a. The entire tensor is flattened into a 1D array and factorised with SVD.

b. Each subject-relation pair is collapsed into a single index along one axis, while the objects remain along the other axis.

c. The subject dimension is repeated until it matches the relation dimension.

d. The relation axis is discarded and the remaining two axes are factorised.

**Correct Answer:** b

**Explanation:**

Unfolding the tensor this way produces a matrix of shape (|E| x |R|) by |E|, where each row is a query of the form (subject, relation, ?) and each column an object. Standard matrix factorisation then applies directly.

(d) would throw away the relation type and conflate distinct facts. (a) loses all structure, which is exactly what the factorisation is meant to exploit.

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**QUESTION 7: [1 mark]**

Which property of hierarchical relations motivates specialised embedding geometries rather than standard Euclidean KG embeddings?

a. Symmetry: (A, is-a, B) implying (B, is-a, A)

b. The frequent presence of cycles in a taxonomy

c. Transitivity: (camel, is-a, mammal) and (mammal, is-a, animal) implying (camel, is-a, animal)

d. The high dimensionality of the entity embeddings

**Correct Answer:** c

**Explanation:**

A taxonomy is a partial order, and its defining property is transitivity. A model such as DistMult scores each triple independently and has no mechanism that makes the third triple plausible once the first two are, so the transitive closure has to be memorised rather than inferred.

(a) is the opposite of what holds - is-a is antisymmetric. (b) is false for a well-formed taxonomy, which is acyclic.

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**QUESTION 8: [1 mark]**

Which statements about hyperbolic (Poincare ball) embeddings for hierarchical data are correct?

a. The volume available grows exponentially with radius, which matches the exponential growth in the number of nodes at each level of a tree.

b. Distances grow slowly near the centre and increase sharply towards the boundary, so depth in a hierarchy maps naturally onto radius.

c. Every node must be placed exactly on the boundary of the unit ball.

d. They give arbitrarily low distortion for any graph, not only tree-like ones.

**Correct Answers:** a, b

**Explanation:**

A tree with branching factor b has roughly b^k nodes at depth k. Euclidean space of fixed dimension cannot accommodate that growth without crowding, which is the source of the distortion. Hyperbolic space expands exponentially with radius, so the tree fits with far less distortion, often in only two dimensions.

(c) is false: points lie strictly inside the ball, since the boundary is at infinite distance. (d) overclaims - the advantage is specific to hierarchical or tree-like structure, and a dense graph with many cycles gains nothing.

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**QUESTION 9: [1 mark]**

Why can a partial-order approach such as order embeddings suit 'is-a' relations better than a purely distance-based one?

a. It encodes the ancestor-descendant relation directly as a coordinate-wise inequality or containment.

b. It guarantees the full transitive closure of the hierarchy is recovered without any training.

c. It represents sibling and ancestor pairs as negatively correlated vectors.

d. It replaces symmetric pairwise distance with an asymmetric notion of ordering, which matches the asymmetry of is-a.

**Correct Answers:** a, d

**Explanation:**

A distance is symmetric by definition, so it can say that two concepts are related but not which one subsumes the other. Order embeddings instead require every coordinate of the child to dominate the corresponding coordinate of the parent, an asymmetric condition that reads off the direction of the relation.

(b) is too strong: the ordering is transitive by construction in the geometry, but the model still has to be trained for the embeddings to actually satisfy the constraints on held-out pairs. (c) is not what these methods do.

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**QUESTION 10: [1 mark]**

Which limitation of axis-aligned cone embeddings do box embeddings address?

a. Cones cannot represent containment at all, whereas boxes can.

b. A cone has the same measure regardless of how broad or narrow it is, so the size of a category cannot be read off its region, whereas a box has a finite volume that can.

c. Cones require all sibling categories to share an apex.

d. Cones cannot be trained with gradient descent.

**Correct Answer:** b

**Explanation:**

An axis-aligned cone extends infinitely, so its measure carries no information about how general or specific the concept is, and the intersection of two cones is degenerate in the same way. That blocks any probabilistic reading of the geometry.

A box is bounded by a lower and an upper corner, so it has a finite volume. Containment along every dimension still encodes is-a, but the volume now reflects the breadth of the category, and the intersection of two boxes is itself a box, which allows partial overlap and conjunction to be represented.

(a) is false - cones do encode containment, that part works fine.

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